Robot Design Log

Check it. Fix it. Fill in the missing parts.

Name: _____ **Date:** _____

This is our team's record of how we built and changed our robot. At judging, the judges will ask us **why** we made each choice and **what we changed**. Some of the answers are written down. Some are missing, and only the people who did the work know them. **That is where you come in.**

What to do

1. **Read** each decision on pages 3 to 12. The gold box shows what we already have written down.
2. **Fix it** if something is wrong. Cross it out and write the right answer beside it.
3. **Fill in** every ✎ question you know the answer to. Put your name next to the ones you worked on.
4. **Add** anything we forgot on pages 14 and 15. Write the test numbers on page 13.
5. **Hand it back** to Chris when you are done. Page 2 is a checklist to tick off.

What makes a good answer?

- **Say why.** "We changed it because ..." beats "We changed it."
- **Use numbers.** "It drove 100 cm and ended 1 cm off" is better than "it was pretty good."
- **Say what did not work.** Judges like honest mistakes. It shows we kept improving.
- **Not sure?** Write what you remember and put a "?" next to it. A guess we can check is better than a blank.

The robot design process we follow

Identify the problem → Design a fix → Create it → Iterate (test and improve) → Communicate what we learned. This log is the proof of that.

Your checklist: what is still missing?

Every decision below is already in our log, but each is missing something. Tick the box when you have filled it in.

#	Decision	What we still need from you	Page	Done
1	**HummerOne** as our base robot	Why, in your own words. Result of the ruler squaring check.	3	☐
2	**Scoring** the four designs	Which score mattered most to you. Any score you would change.	3	☐
3	**Mine cart wheels** at the front	Who changed them and when. Why this wheel.	4	☐
4	**Second colour sensor**	Where it is mounted and why. Proof that squaring works now.	4	☐
5	**Mini ball caster** plan	Why the "mini" size. Any drifting when it was tested.	5	☐
6	**Attachments** on the mounting bar	Pros and cons of all three attachments, and a rating for each.	5	☐
7	**Quick pins** instead of axle pins	The date. What Jack hoped axle pins would do. Other attachments checked.	6	☐
8	**Gyro PID** drive and turn blocks	Mission 2 test results, in your own words.	6	☐
9	**Technic ball joint** and retuning	Which part it is. What the wobble looked like.	7	☐
10	Distance in **cm**	The distances used in the Mission 2 code. Repeat 1 m checks.	7	☐
11	**Seed basket** for Mission 11	Ideas tried and catches out of 10. How it relates to the seed bucket.	8	☐
+	**Final attachment pick**	Which one we chose for final refinement, and why.	8	☐
12	**Forklift** vs the original build	What we changed from the original, and why. Tipping test.	9	☐
13	**Seed bucket** vs the original lifting arm	What we changed, the seed order, how seeds come out.	10	☐
14	**Mission 1** attachment vs the original	Which original build it is, and what we changed.	11	☐
15	**2-axis (with hook)** vs the original	What we changed, and what the hook does.	12	☐
+	**Test results**	Drive 1 m ten times, ten 90 degree turns, ruler squaring check.	13	☐
+	**Anything we forgot**	Other changes to the robot, attachments or code.	14-15	☐

Why do judges care about this?

The judges look for a robot design we can explain, evidence that we tested it more than once, and changes we made because of what the tests showed. Every blank you fill in is one more answer we can give out loud.

1 We chose HummerOne as our base robot

Decided 2026-09-22

WHAT OUR LOG SAYS

We shortlisted four designs: HummerOne, Comp Bot, StarterBot, and GummyBears / Droid Bot M. We picked HummerOne because it is free, has a full-width attachment bar (swaps in about 5 seconds, versus about 20 for a pinned design), and its frame should square well against a wall. It has 73 build steps. We chose not to buy StarterBot ($30). The build was finished on 2026-10-01.

✎ **In your own words, why HummerOne and not one of the others?** (Pick the reason that matters most to you.)

✎ **What do you remember about building it?** (Surprises, problems, how long it took, who helped.)

✎ **Ruler squaring check: does the body touch the ruler before any wheel does?** (Watch the front-left wheel. Put the numbers on page 13.)

2 We scored the four designs

2026-09-22, whole team

WHAT OUR LOG SAYS

Six criteria, each scored 1 to 5, all about how the robot performs in a 2.5 minute match: archetype match, attachment swap speed, squaring, stability, maneuverability, and drive repeatability. **Totals out of 30: HummerOne 23, Comp Bot 20.5, GummyBears / Droid Bot M 20, StarterBot 14.5.** Cost and build time were left out on purpose and decided separately.

✎ **Which of the six criteria mattered most to you, and why?**

✎ **Was there a score you disagreed with?** (Which design, which criterion, and what would you give it?)

✎ **Why did we leave cost out of the scoring?**

3 Front wheels: mine cart wheels

Before 2026-10-01

WHAT OUR LOG SAYS

HummerOne's front wheels have no rubber, so they skid unevenly from run to run. We swapped in mine cart wheels, which are lower and create less friction when turning. The conference taught us that smaller wheels wobble less and that **accuracy matters more than speed**. Problem: the new wheels still stuck out past the sides, so a wheel could touch a wall before the body. They were good in a straight line but resisted turning. We replaced them in entry 9.

Who changed the wheels? _____ **Date:** _____

✎ **Why mine cart wheels and not a different wheel?** (What else did you try or look at?)

✎ **How did you notice they stuck out past the sides?** (What did you see or test?)

4 We added a second colour sensor

2026-09-27 to 09-29, Jack

WHAT OUR LOG SAYS

At the conference we learned that squaring up on a line needs **two** colour sensors: when one sees black, the robot turns until the other sees black too, then wiggles straight. In the gym we could not square up with one sensor, so Jack added a second. The square-up program was finished on 2026-10-01. Still to do: find which lines in front of mission models we can square up on, and show a judge a before and after.

✎ **Where is the second sensor mounted, and why there?**

✎ **Before and after: what happened when the robot was placed crooked, without and with squaring?** (Write or draw what you saw.)

✎ **Which mission models have a line in front we could square up on?**

5 Plan: mini ball caster at the front

Fedor and Ira, 2026-09-29

WHAT OUR LOG SAYS

A ball caster has almost no friction, turns best, and is small. It sits **below** the body instead of on the sides, so the flat body faces can touch a wall first. It was postponed on 2026-10-01 because Fedor and Ira were sick. It was finished on 2026-10-13 with a Technic ball joint (entry 9). Known risk: a caster can drift on hard acceleration and when reversing.

✎ **Why did we choose the mini size?** (Ask Fedor or Ira if you are not sure.)

✎ **Did it drift when driving or reversing?** (What did you test, and what happened?)

6 Mission 1 attachments on HummerOne's mounting bar

Status 2026-10-01

WHAT OUR LOG SAYS

Plan: adapt three Mission 1 attachments to the HummerOne bar, test all three, rate the pros and cons, then pick one for final refinement. **Done:** lifting arm with a seed bucket, and 2-axis attachment with a hook. **In progress:** forklift (it is the tallest when extended, so test it for tipping). Ira is writing down the seed order the mission needs.

✎ **Rate each attachment and tell us the pros and cons.** (What we changed from each original build is on pages 9 to 12.)

Attachment	Pros	Cons	Rating 1 to 5
Lifting arm + seed bucket			
2-axis + hook			
Forklift			

✎ **Did the forklift tip over when raised?** (What did you try to fix it?)

✎ **Who built each attachment, and what did you change from the original video?**

7 Attachment pins: axle pins tried, quick pins kept

Tested 2026-10-01

WHAT OUR LOG SAYS

Jack suggested axle pins as an alternative to quick pins. While testing the lifting arm with the seed bucket, the attachment kept coming off with axle pins. We went back to **quick pins, which hold more securely**. A secure fit matters more than swap speed, because an attachment that falls off mid-run loses points and time.

When did Jack suggest it? _____ **Date:** _____

✎ **What did Jack hope axle pins would do better?** (Faster swap? Smoother? Something else?)

✎ **Did the 2-axis (hook) and the forklift also come off with axle pins? How long does a quick pin swap take?**

8 Gyro PID drive and turn blocks (Mission 2 code)

2026-10-06, Jordan and Fedor

WHAT OUR LOG SAYS

Yaw (gyro) turns the robot by a real angle, but 90 motor degrees only turns a wheel a quarter turn. **Gyro Move** drives straight by fixing its heading error every 0.1 seconds (Kp 2.1, Ki 2), with a speed ramp. **Gyro Turn** slows as it nears the angle and stops 3 degrees early so it does not overshoot. Both are reusable My Blocks. Used in a Mission 2 test run on 2026-10-06.

✎ **How did the Mission 2 test runs go?** (What worked, what failed, what to change?)

How Kp and Ki were tuned, and the tests for each attachment, are recorded in the separate Kp/Ki test document.

✎ **In one sentence, why is gyro driving better than motor degrees?**

9 Technic ball joint instead of mine cart wheels

Weekend before 2026-10-13, Jordan

WHAT OUR LOG SAYS

The mine cart wheels were very good in a straight line but resisted turning and stuck out past the sides. The Technic ball joint is better for turns and stays under the robot, so the flat sides can touch a wall. Side effect: the robot **wobbled** when driving straight. Jordan adjusted Kp and Ki in Gyro Move until it went straight again. On 2026-10-06 the values were Kp 2.1 and Ki 2.

Which Technic ball joint part is it? (name or part number) _____

The new Kp and Ki values go in the separate Kp/Ki test document.

✎ **Describe the wobble. How did you know it was fixed?**

✎ **Does the ball joint drift when accelerating hard or reversing?**

10 Gyro Move takes distance in centimetres

Weekend before 2026-10-13, Jordan

WHAT OUR LOG SAYS

Distance in cm is easier to communicate and test: measure on the table with a ruler and type it in. We drove a measured **100 cm with a 1 m measuring stick, and it took 2070 motor degrees**, so Gyro Move uses **20.7 motor degrees per cm**. The Mission 2 distances (140 and 625 motor degrees) were converted to about 6.8 cm and 30.2 cm.

✎ **What distances (in cm) are really used in the Mission 2 code now?**

Move 1: _____ **cm Move 2:** _____ **cm Other:** _____

✎ **Why measure on the table instead of working it out from the wheel size?** (Hint: think about tyre squash and slip.)

✎ **Did you repeat the 1 m drive?** (Write the results on page 13.)

11 Mission 11 seed-catching basket

2026-10-13, Jordan, Brynn, Jack

WHAT OUR LOG SAYS

We built a basket attachment to catch the seed on Mission 11. It catches the seed, but **sometimes the seed goes over the top**. Ideas to try: higher walls, a lip or a lid, or a slower catch. Jordan is also writing the Mission 11 code.

✎ **What ideas did you try to stop the seed going over? How many catches out of 10 did each get?**

Idea we tried	Catches out of 10	What happened

✎ **How does the basket relate to the seed bucket on the lifting arm (entry 6)?** (Same attachment, or different?)

★ Final attachment pick

Still to record

WHAT OUR LOG SAYS

We plan to test all three Mission 1 attachments on HummerOne, rate their pros and cons (page 5), and pick one for final refinement. The reasons for the pick are not recorded yet.

✎ **Which attachment do you think we should pick for final refinement?**

✎ **Why?** (Use your ratings and test results. What made it better than the other two?)

✎ **What would you improve about it next?**

12 Forklift

Jack, Ira, Ella, Amelia

WHAT OUR LOG SAYS

Original build: **Forklift1-zoptyk.pdf**, a 73 page build in 3 phases that lifts a platform with rack and pinion. It was Ella's first-to-implement task from Robot Man ("raise the platform"). We are adapting it to HummerOne's mounting bar. Our review warned it is the **tallest when extended** and the **biggest centre of gravity risk**, so we need to test it for tipping. Jack and Ira worked on it 2026-10-06, Ella and Amelia 2026-10-13. Finish target was 2026-10-15.

✎ **Original build** (draw it or tape a photo)

✎ **Our version** (draw it or tape a photo)

✎ **What did we change from the original build, and why?** (One block per change. Need more room? Use the back of the page.)

Change 1 · Part or area: Mounting

The original build had:

We changed it to: Adapted to HummerOne's mounting bar

Why?

Change 2 · Part or area:

The original build had:

We changed it to:

Why?

Change 3 · Part or area:

The original build had:

We changed it to:

Why?

✎ **How high does it lift, and does the robot tip when it is raised?** (Say what you tried to fix it.)

✎ **Which mission does it do, and what is it carrying or lifting?**

Kp and Ki testing for this attachment is recorded in the separate Kp/Ki test document, not here.

13 Seed bucket (on the lifting arm)

Sylvie, Ella, Alexis, Amelia, Brynn

WHAT OUR LOG SAYS

Original build: **Lifting-Arm-Attachment.pdf**, a 15 page geared hinge arm for scoop and hook missions. Our review warned about the **forward centre of gravity shift**. We added a **seed bucket**. The seed mission needs the seeds to go in a **specific order**, which Ira is writing down. With axle pins the arm kept coming off, so we switched back to **quick pins** (entry 7).

Original build (draw it or tape a photo)

Our version (draw it or tape a photo)

What did we change from the original build, and why? (One block per change. Need more room? Use the back of the page.)

Change 1 · Part or area: Arm end

The original build had: Plain geared hinge arm

We changed it to: Added a seed bucket

Why?

Change 2 · Part or area: Pins

The original build had:

We changed it to: Quick pins (axle pins kept coming off)

Why? Holds more securely

Change 3 · Part or area:

The original build had:

We changed it to:

Why?

What is the correct seed order, and how does the bucket keep the seeds in order?

How do the seeds get out of the bucket, and does it work every time?

Kp and Ki testing for this attachment is recorded in the separate Kp/Ki test document, not here.

14 Mission 1 attachment

Three options, one group each

WHAT OUR LOG SAYS

We built three Mission 1 options, inspired by **Giraffes #3388** (Amelia, Ella), **Platinum Pizzeria** (Fedor, Ira) and **Team TurtleBots** (Jordan, Alexis). The plan is to adapt each to the HummerOne bar, rate the pros and cons, and pick one. **Not confirmed yet:** which original build this page is about. Our review suggested **Multipurpose-Single-Attachment.pdf**, a 9 page paddle or flipper, as the fastest route to a working Mission 1 tool.

✎ **Original build** (draw it or tape a photo)

✎ **Our version** (draw it or tape a photo)

✎ **What did we change from the original build, and why?** (One block per change. Need more room? Use the back of the page.)

Change 1 · Part or area: _____

The original build had: _____

We changed it to: _____

Why? _____

Change 2 · Part or area: _____

The original build had: _____

We changed it to: _____

Why? _____

Change 3 · Part or area: _____

The original build had: _____

We changed it to: _____

Why? _____

Which original build is this? _____

Who built it? _____

✎ **What does Mission 1 need the attachment to do, and does it do it every time?**

Kp and Ki testing for this attachment is recorded in the separate Kp/Ki test document, not here.

15 2-axis movement (with hook)

Status 2026-10-01

WHAT OUR LOG SAYS

Original build: **2-Axis-Movement-Attachment.pdf**, a 13 page build where bevel gears give **vertical and horizontal motion from one input**, so one attachment can do two mission actions without a swap. Our review warned that **gear backlash reduces precision**. We modified it with a **hook**.

Original build (draw it or tape a photo)

Our version (draw it or tape a photo)

What did we change from the original build, and why? (One block per change. Need more room? Use the back of the page.)

Change 1 · Part or area: End of arm

The original build had: No hook

We changed it to: Added a hook

Why?

Change 2 · Part or area:

The original build had:

We changed it to:

Why?

Change 3 · Part or area:

The original build had:

We changed it to:

Why?

Which mission does the hook do, and what does it hook onto?

Do the gears slip or have play (backlash)? How did you handle it?

Kp and Ki testing for this attachment is recorded in the separate Kp/Ki test document, not here.

Test results: our evidence

Judges look for proof that we tested the robot **more than once**. Fill in what we have measured. If a test has not been done yet, do it, then write the numbers here. Kp and Ki tests for each attachment go in the separate Kp/Ki test document.

A. Drive 1 m (100 cm), ten times

Run	Ended at (cm)	Off by (cm)
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		

B. Turn 90 degrees, ten times

Run	Turned to (deg)	Off by (deg)
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		

C. Ruler squaring check (Hold a ruler flat against each face. The body should touch before any wheel does. Watch the front-left drive wheel.)

Face	Body touches first? Yes / No	Notes (which wheel touched first, by how much)
Back		
Left side		
Right side		

✎ **What do these numbers tell us?** (Is the robot repeatable? What would you change because of them?)

Tested by: _____ **Date:** _____

Did we forget something?

Think about anything you changed or decided that is not in the log: a wheel, a gear, a sensor, a part of the frame, an attachment, a program. Write each one down so the judges can hear the whole story.

16 Decision: _____

Who: _____ **Date:** _____

What else did we think about doing?

Why did we choose this one? (What problem did it solve? What did we see or measure?)

What happened? What is still left to do?

17 Decision: _____

Who: _____ **Date:** _____

What else did we think about doing?

Why did we choose this one? (What problem did it solve? What did we see or measure?)

What happened? What is still left to do?

18 Decision: _____

Who: _____ **Date:** _____

What else did we think about doing?

Why did we choose this one? (What problem did it solve? What did we see or measure?)

What happened? What is still left to do?

19 Decision: _____

Who: _____ **Date:** _____

What else did we think about doing?

Why did we choose this one? (What problem did it solve? What did we see or measure?)

What happened? What is still left to do?

✓ One more thing

Something you are proud of about our robot, and one thing you would still like to improve.